Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters

Research Paper — Exploratory Data Analysis, Insights & Recommendations

Dataset: Transaction-level sales data, FY 2025

Prepared by: *Deepika Dohare*

---

1. Introduction and Background

In the specialty coffee and beverage retail industry, understanding product performance is essential for effective inventory management, menu planning, pricing decisions, and revenue growth. Businesses offer a wide range of products, but not every product contributes equally to sales or revenue. Some products generate high sales volumes, while others contribute more revenue despite relatively lower demand.

Without structured analysis, decisions regarding product promotion, inventory allocation, and menu optimization may rely heavily on assumptions rather than measurable business performance.

Afficionado Coffee Roasters operates across multiple store locations and offers a diverse portfolio of coffee, tea, bakery, and related products. This project examines transaction-level sales data to understand which products and categories generate the greatest revenue, how sales performance varies across stores, and whether product popularity corresponds with revenue contribution.

The objective is to transform historical sales data into meaningful business insights that can support more informed product and operational decisions.

1.1 Problem Statement

The availability of transaction-level sales data does not automatically provide clear visibility into product performance. Without systematic analysis, the business may find it difficult to identify its strongest revenue contributors, recognize products with lower performance, understand category-level revenue dependency, and evaluate differences in store performance.

This project addresses these challenges through exploratory data analysis and interactive visualization.

1.2 Project Objectives

The primary objectives are to:

- Identify leading products based on sales volume and revenue generation.
- Measure revenue contribution across products and product categories.
- Examine the relationship between product popularity and revenue.
- Compare revenue performance across store locations.
- Identify peak sales hours and understand sales timing patterns.
- Evaluate revenue concentration across the product portfolio.
- Develop data-driven recommendations for product and operational optimization.

2. Dataset Description

The analysis uses transaction-level sales data for FY 2025. The dataset contains **149,116 transaction records**, covering 214,470 units sold across 80 individual products, 9 product categories, and 3 store locations.

Revenue was calculated using the quantity purchased and the corresponding unit price for each transaction.

Dataset overview

Metric	Value
Transaction records	149,116
Total units sold	214,470
Unique products	80
Product categories	9
Store locations	3
Total revenue	698,812.33
Reporting year	2025

The dataset includes transaction identifiers, transaction time, quantity, unit price, store information, product identifiers, product categories, product types, and product details.

Data validation was performed to check missing values, duplicate transaction identifiers, and invalid quantity or price values. The analysis dataset contained no missing values, duplicate transaction identifiers, or non-positive quantities and prices.

One important limitation is that the dataset does not contain product cost, profit margin, customer identifiers, or transaction dates at daily or monthly granularity. Consequently, the analysis focuses on sales revenue and operational patterns rather than profitability, customer behaviour, or seasonal trends.

3. Analytical Methodology

The project followed a structured exploratory data analysis process using Python.

Data preparation and validation: The dataset was loaded and inspected using Pandas. Data types, missing values, duplicate records, and invalid numerical values were checked to establish data reliability.

Revenue computation: Revenue was calculated at the transaction level using:

Revenue = Transaction Quantity × Unit Price

The resulting revenue values were aggregated across products, product categories, product types, and store locations.

Product performance analysis: Products were ranked separately by total units sold and total revenue. This enabled identification of high-volume products, high-revenue products, and differences between popularity and revenue contribution.

Category analysis: Revenue and sales volume were aggregated by product category to understand the relative importance of each category within the overall product portfolio.

Store performance analysis: Revenue, units sold, and transaction records were compared across the three store locations to identify differences in their contribution.

Time-based analysis: Transaction times were used to examine hourly sales patterns and identify periods of higher sales activity.

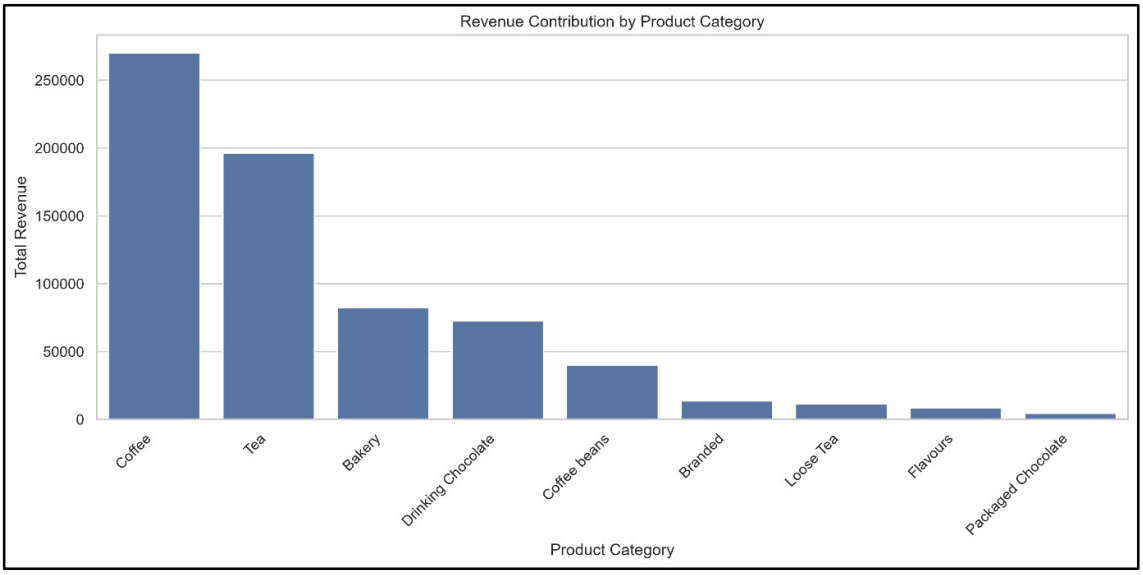
Revenue concentration analysis: A Pareto analysis was conducted by ranking products according to revenue and calculating cumulative revenue contribution. This helped identify the number of products responsible for approximately 80% of total revenue.

Product performance classification: Products were classified using median sales volume and median revenue to distinguish high-volume/high-revenue products from other performance groups.

The analysis was performed using Python, Pandas, NumPy, Matplotlib, Seaborn, and Plotly. An interactive Streamlit application was subsequently developed to make the findings accessible through visualizations and filters.

4. Key Findings and Business Insights

4.1 Category Revenue Contribution



The analysis identified Coffee and Tea as the two largest revenue-generating categories.

Coffee contributed **269,952.45 (38.63%)** of total revenue, while Tea contributed **196,405.95 (28.11%)**. Together, these categories accounted for **66.74% of total revenue**.

Bakery contributed 11.78%, followed by Drinking Chocolate at 10.36%. The remaining categories individually contributed smaller shares.

This distribution highlights the importance of Coffee and Tea within the business portfolio. Their strong contribution represents a significant commercial strength, while also indicating potential exposure if demand in either category declines.

4.2 Leading Products by Revenue and Sales Volume

At the individual product level, Sustainably Grown Organic Lg emerged as the highest revenue-generating product, contributing 21,151.75, or approximately 3.03% of total revenue.

It was followed by Dark chocolate Lg at 21,006.00, Latte Rg at 19,112.25, Cappuccino Lg at 17,641.75, and Morning Sunrise Chai Lg at 17,384.00.

However, the highest revenue-generating product was not necessarily the most popular by sales volume. Earl Grey Rg recorded the highest sales volume, with 4,708 units sold.

This difference demonstrates why product performance should be evaluated using both sales volume and revenue rather than relying on a single indicator.

4.3 Popularity Versus Revenue Performance

Comparing product popularity with revenue contribution revealed differences in how products perform commercially.

Some products achieve strong sales volumes but generate relatively lower revenue, while others contribute substantial revenue despite selling fewer units.

The product performance classification, based on median sales volume of 3,629.5 units and median product revenue of 9,225.00, identified four groups:

- **High Volume – High Revenue:** 38 products (47.5%), representing strong overall performers.
- **Low Volume – Low Revenue:** 38 products (47.5%), representing products that may require further performance evaluation.
- **Low Volume – High Revenue:** 2 products (2.5%), representing lower-volume products with relatively strong revenue contribution.
- **High Volume – Low Revenue:** 2 products (2.5%), representing products with strong demand but comparatively lower revenue.

This classification provides a practical framework for identifying products that may benefit from promotion, pricing review, improved availability, or further evaluation.

4.4 Store Performance

The three store locations demonstrated relatively balanced revenue performance.

Hell's Kitchen generated the highest revenue at **236,511.17 (33.84%)**, followed by Astoria at 232,243.91 (33.23%) and Lower Manhattan at 230,057.25 (32.92%).

The relatively small difference between the three locations suggests that revenue contribution is distributed fairly evenly across the store network.

This finding provides an opportunity to examine store-specific product preferences and operational practices without assuming that one location significantly underperforms the others.

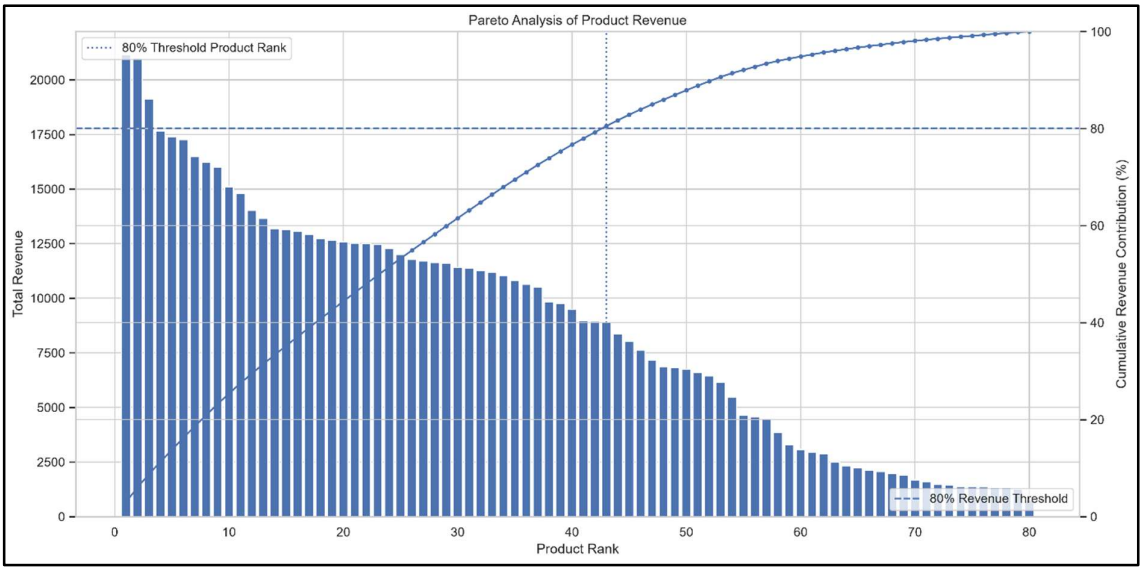
4.5 Peak Sales Hour Analysis

The hourly analysis identified **10:00 AM as the highest-revenue hour**, generating 88,673.39, equivalent to 12.69% of total revenue.

The 8:00 AM to 10:00 AM period also demonstrated strong sales activity, highlighting the importance of morning operations.

This insight can support better staff scheduling, inventory preparation, and product availability during higher-demand periods.

4.6 Revenue Concentration and Pareto Analysis



The Pareto analysis examined cumulative revenue contribution across all 80 products.

The results indicated that **43 products generated approximately 80.52% of total revenue**. The top 20 products contributed 44.38%, while the top 40 contributed 76.68%.

Unlike a highly concentrated portfolio in which only a few products generate most revenue, this result indicates that a relatively broad group of products contributes meaningfully to the business.

The remaining products still represent an opportunity for further evaluation, but low revenue contribution alone should not be treated as sufficient evidence for discontinuation. Products may also serve strategic, complementary, or customer-choice purposes.

5. Recommendations

Based on the analytical findings, the following recommendations are proposed:

1. Prioritize High-Performing Products: Maintain consistent availability of leading revenue-generating products and high-volume items. Use product rankings to support inventory allocation, menu placement, and promotional planning.

2. Strengthen Core Category Performance: Coffee and Tea account for 66.74% of revenue. The business should protect the performance of these categories while exploring opportunities to develop other categories and diversify revenue contribution.

3. Review Products with Different Volume and Revenue Patterns: Evaluate high-volume, relatively low-revenue products for pricing and product positioning opportunities. Similarly, consider targeted promotion for lower-volume products with strong revenue contribution.

4. Optimize Store-Level Decisions: Although store revenues are relatively balanced, product-level comparisons across locations may help identify opportunities for location-specific inventory planning and product assortment.

5. Improve Peak-Hour Operations: Prepare staffing and inventory around the strong morning sales period, particularly between 8:00 AM and 10:00 AM, while monitoring hourly demand over time.

6. Conduct Regular Product Portfolio Reviews: Use the revenue concentration analysis and product performance classification to identify products requiring closer attention. Any menu reduction decision should also consider costs, margins, customer preferences, and operational requirements.

6. Conclusion

This project demonstrates how transaction-level sales data can be transformed into meaningful business intelligence for product and revenue optimization at Afficionado Coffee Roasters.

The analysis identified Coffee and Tea as the primary revenue-generating categories, highlighted leading products by revenue and sales volume, established that store performance is relatively balanced, and identified 10:00 AM as the peak revenue hour. The Pareto analysis further showed that 43 out of 80 products contributed approximately 80.52% of total revenue.

These findings provide a structured basis for inventory planning, product prioritization, category management, store-level evaluation, and operational improvements.

An interactive Streamlit dashboard was also developed to present the findings through key performance indicators, product rankings, category revenue visualizations, store comparisons, popularity-versus-revenue analysis, and product-level drill-down capabilities. Filters for store location, product category, and product type allow users to explore performance from different perspectives.

While the project provides useful insights into sales and revenue performance, future analysis could be strengthened by incorporating product costs, profit margins, customer-level information, and more detailed transaction dates. These additions would enable profitability analysis, customer segmentation, and seasonal demand forecasting.

Overall, the project establishes a data-driven foundation for improving product management, supporting operational efficiency, and identifying future revenue growth opportunities.

End of Research Paper